

Lecture 4

Kinetics of Particles

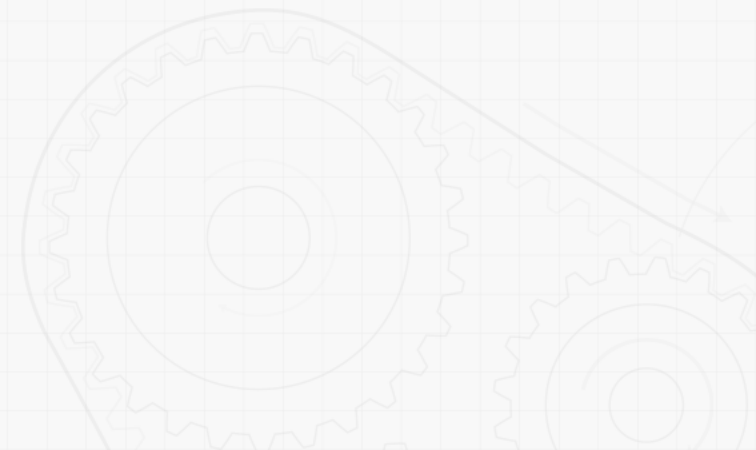
Prof. Pilwon Hur
MC2103
Fall 2026



4 Kinetics of Particles: Newton's Second Law

Kinetics of Particles

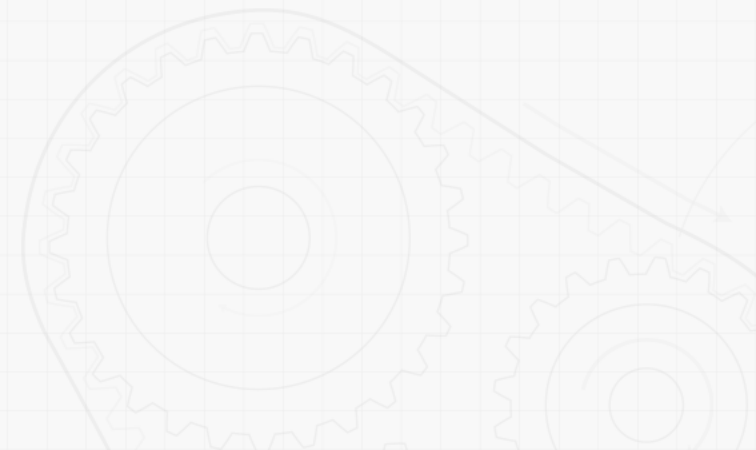
We must analyze all of the forces acting on the racecar in order to design a good track.



4 Kinetics of Particles: Newton's Second Law

Kinetics of Particles

As a centrifuge reaches high velocities, the arm will experience very large forces that must be considered in design.



4 Kinetics of Particles: Newton's Second Law

Introduction

⚙️ Newton's Second Law of Motion

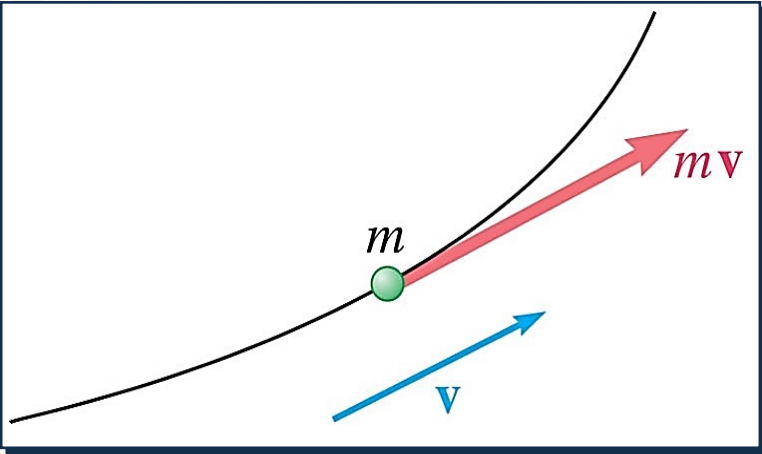
- If the resultant force acting on a particle is not zero, the particle will have an acceleration proportional to the magnitude of resultant and in the direction of the resultant.

$$\sum F = ma$$

- Must be expressed with respect to a *Newtonian (or inertial) frame of reference*, i.e., one that is not accelerating or rotating.
- This form of the equation is for a **constant mass system**.

4 Kinetics of Particles: Newton's Second Law

Linear Momentum of a Particle



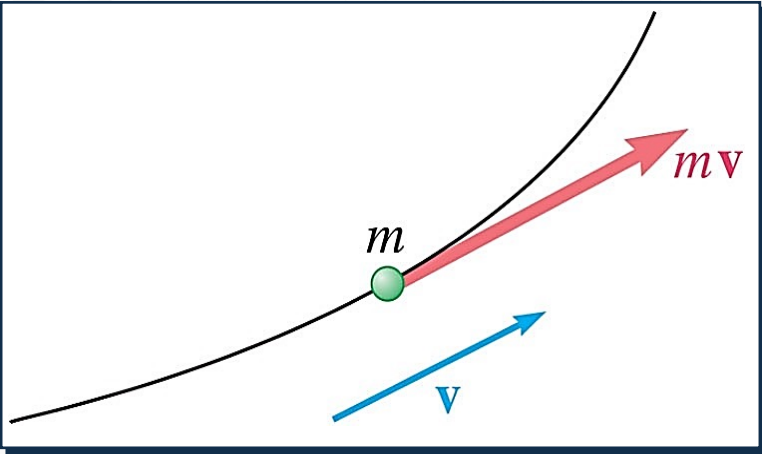
- Replacing the acceleration by the derivative of the velocity yields,

$$\begin{aligned}\sum \vec{F} &= m \frac{d\vec{v}}{dt} \\ &= \frac{d}{dt} (m\vec{v}) = \frac{d\vec{L}}{dt}\end{aligned}$$

$\vec{L}$ = linear momentum of the particle

4 Kinetics of Particles: Newton's Second Law

Linear Momentum of a Particle



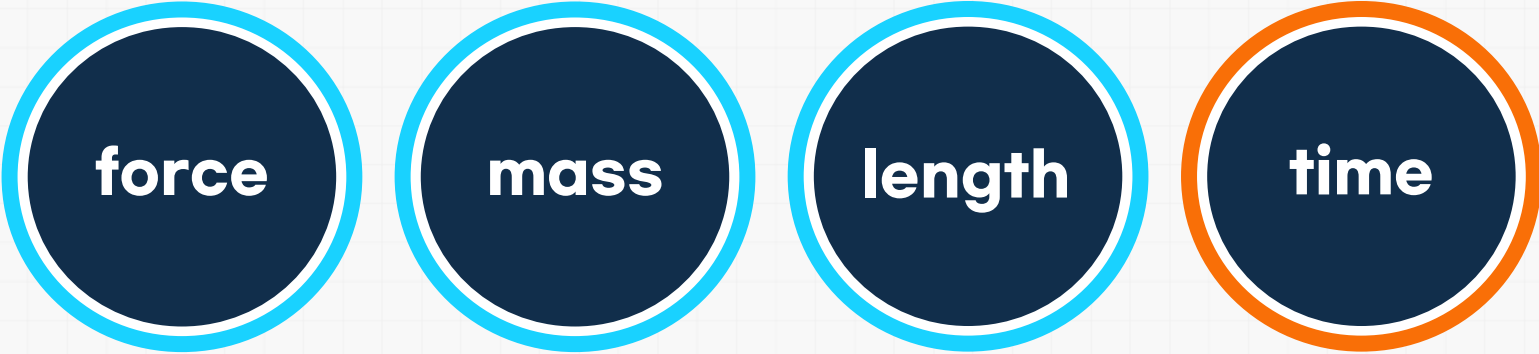
- *Linear Momentum Conservation Principle:*
 - If the resultant force on a particle is **zero**, the linear momentum of the particle **remains constant** in both **magnitude and direction**.

4 Kinetics of Particles: Newton's Second Law

Systems of Units

⚙️ Primary dimensions

- Of the units for the four primary dimensions, three may be chosen arbitrarily.
- The fourth must be compatible with **Newton's 2nd Law**.



force

mass

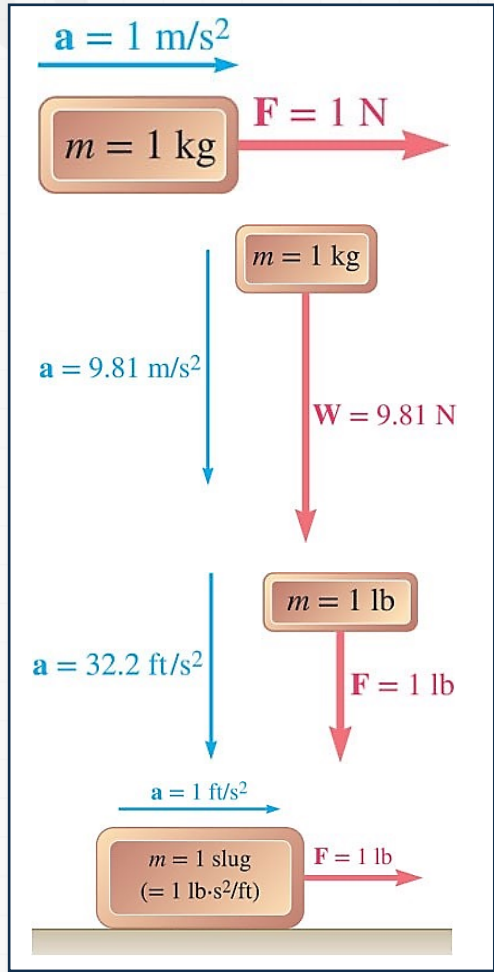
length

time



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Systems of Units



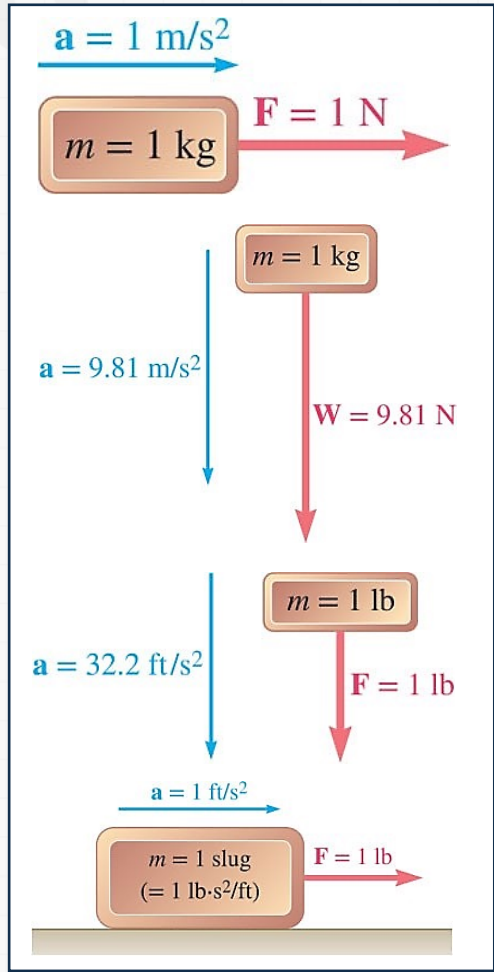
- *International System of Units (SI Units):*

- Base units are the units of length (m), mass (kg), and time ($second$).
- The unit of force is derived,

$$1\text{N} = (1\text{kg})(1 \frac{\text{m}}{\text{s}^2}) = 1 \frac{\text{kg}\cdot\text{m}}{\text{s}^2}$$

4 Kinetics of Particles: Newton's Second Law

Systems of Units



- *U.S. Customary Units:*

- Base units are the units of force (*lb*), length (*m*), and time (*second*).
- The unit of mass is derived,

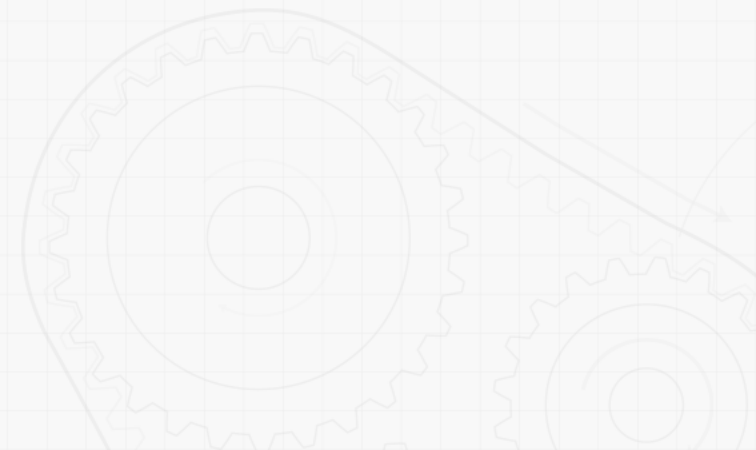
$$1 \text{ lbm} = \frac{1 \text{ lb}}{32.2 \text{ ft/s}^2} \quad 1 \text{ slug} = \frac{1 \text{ lb}}{1 \text{ ft/s}^2} = 1 \frac{\text{lb} \cdot \text{s}^2}{\text{ft}}$$

4 Kinetics of Particles: Newton's Second Law

Equations of Motion

⚙ Newton's second law

$$\sum F = ma$$

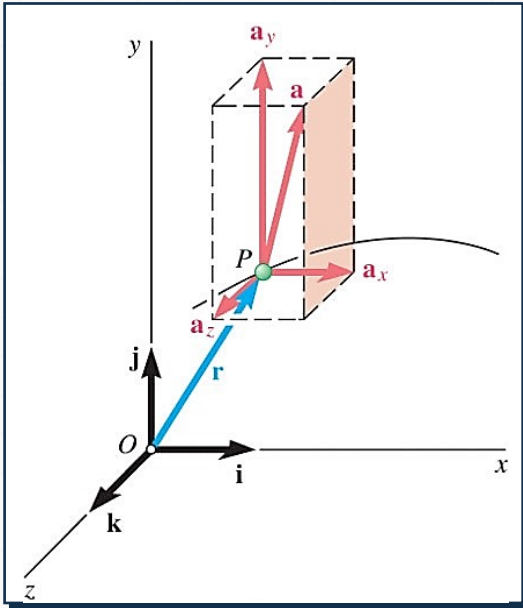


4 Kinetics of Particles: Newton's Second Law

Equations of Motion

⚙ Newton's second law

- Can use scalar component equations, example for rectangular components,



$$\Sigma(F_x \hat{i} + F_y \hat{j} + F_z \hat{k}) = m(a_x \hat{i} + a_y \hat{j} + a_z \hat{k})$$

$$\Sigma F_x = ma_x \quad \Sigma F_y = ma_y \quad \Sigma F_z = ma_z$$

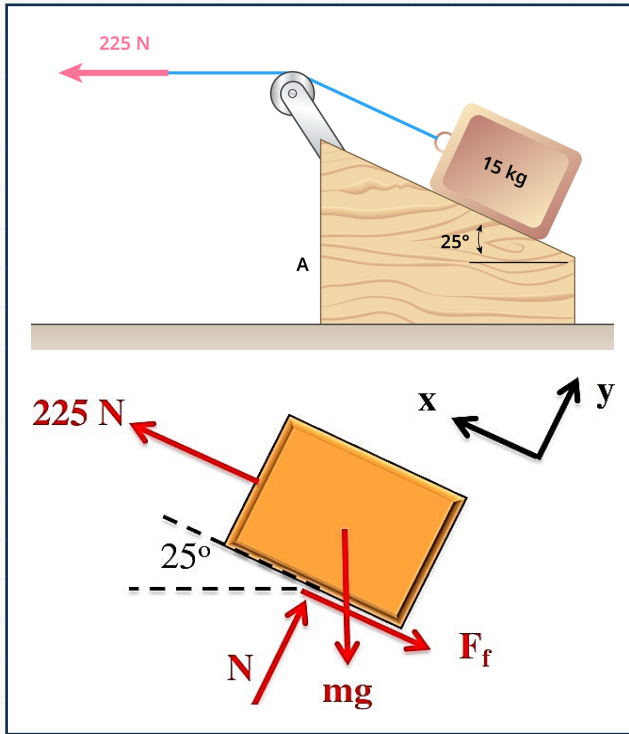
$$\Sigma F_x = m\ddot{x} \quad \Sigma F_y = m\ddot{y} \quad \Sigma F_z = m\ddot{z}$$

4 Kinetics of Particles: Newton's Second Law

Free Body Diagrams and Kinetic Diagrams

⚙ The free body diagram

- The free body diagram is the same as you have done in statics; we will add the kinetic diagram in our dynamic analysis.



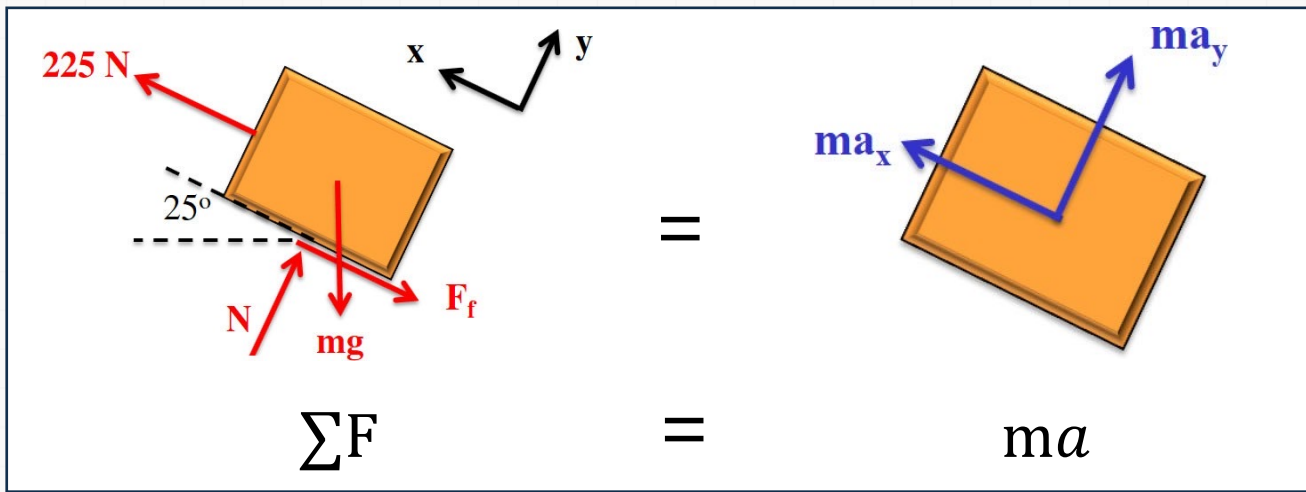
- Isolate **the body of interest** (free body).
- Draw your **axis system** (example: Cartesian, polar, path).
- Add in **applied forces** (example: weight, 225 N pulling force).
- Replace supports with **forces** (example: normal force).
- Draw appropriate **dimensions** (usually angles for particles).

4 Kinetics of Particles: Newton's Second Law

Free Body Diagrams and Kinetic Diagrams

⚙️ Put the inertial terms for the body of interest on the kinetic diagram.

- Isolate **the body of interest** (free body).
- Draw in the mass times acceleration of the particle; if unknown, do this in the **positive direction according to your chosen axes**.

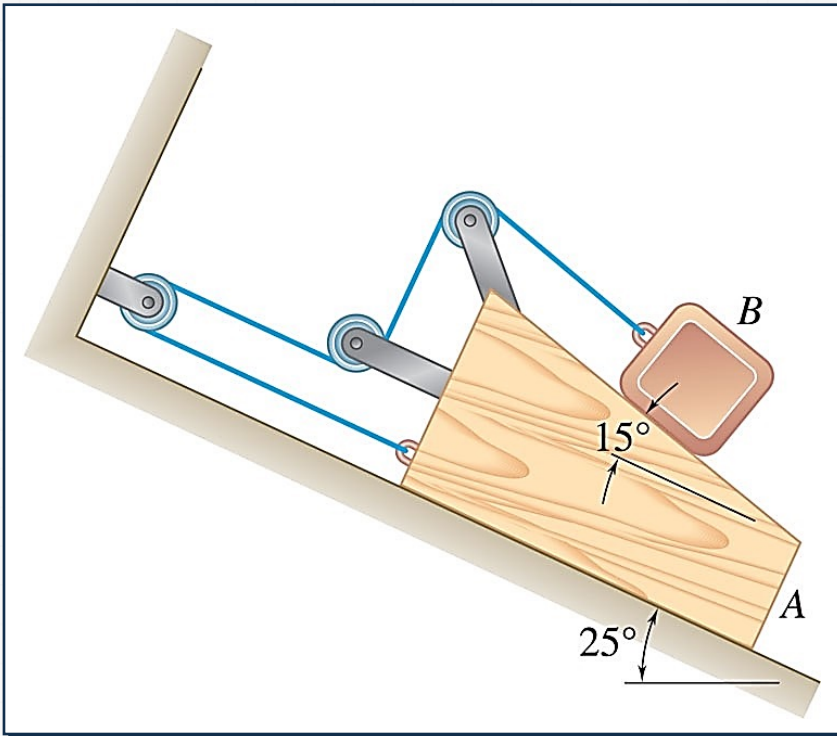


4 Kinetics of Particles: Newton's Second Law

Free Body Diagrams and Kinetic Diagrams

⚙ Draw the FBD and KD for block A

- (note that the massless, frictionless pulleys are attached to block A and should be included in the system)

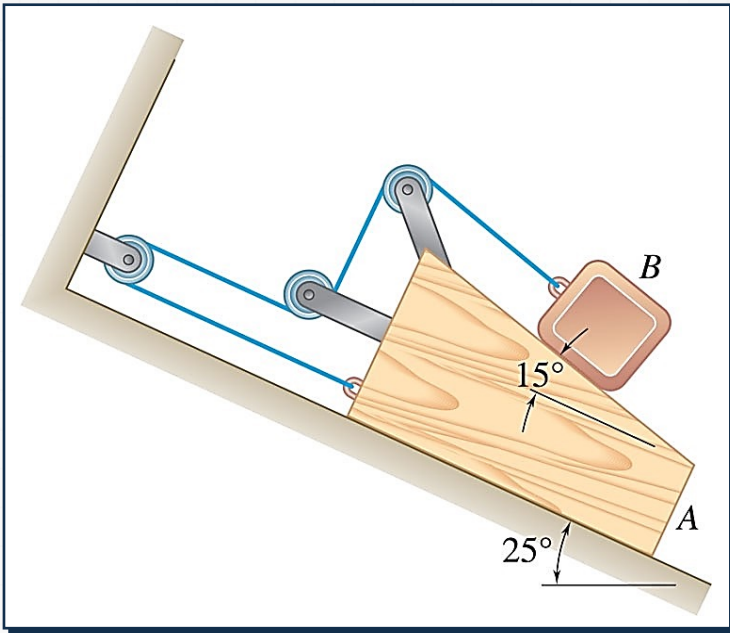


4 Kinetics of Particles: Newton's Second Law

Free Body Diagrams and Kinetic Diagrams

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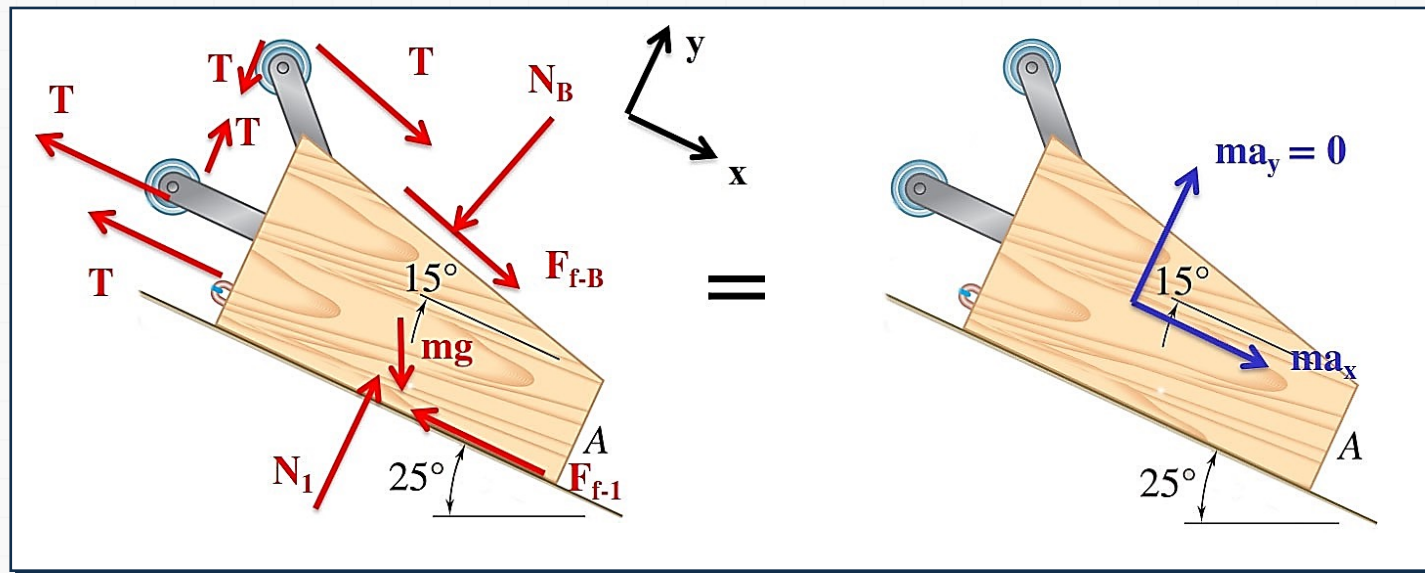


- Isolate body
- Axis
- Applied forces
- Replace supports with forces
- Dimensions (already drawn)
- Kinetic diagram

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Free Body Diagrams and Kinetic Diagrams

⚙ Draw the FBD and KD for block A

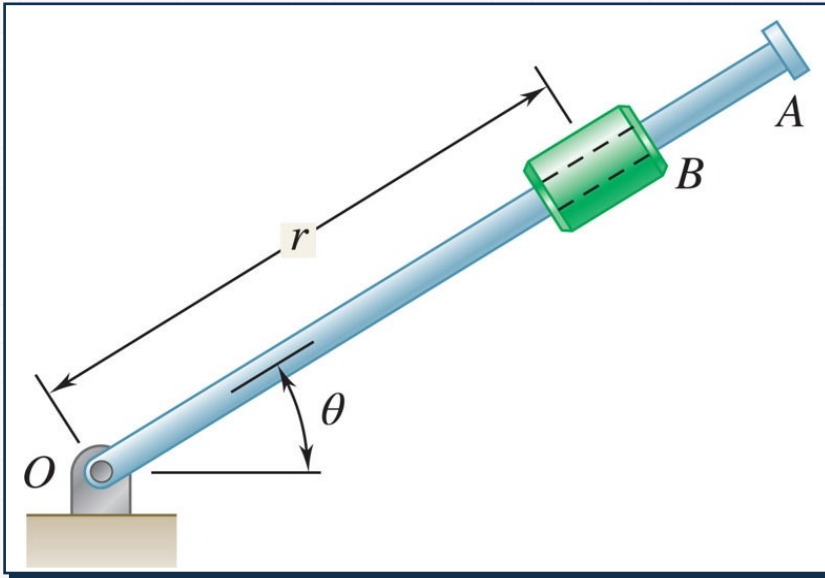


4 Kinetics of Particles: Newton's Second Law

Free Body Diagrams and Kinetic Diagrams

⚙ Draw the FBD and KD for the collar B.

- Assume there is friction acting between the rod and collar, motion is in the vertical plane, and θ is increasing.

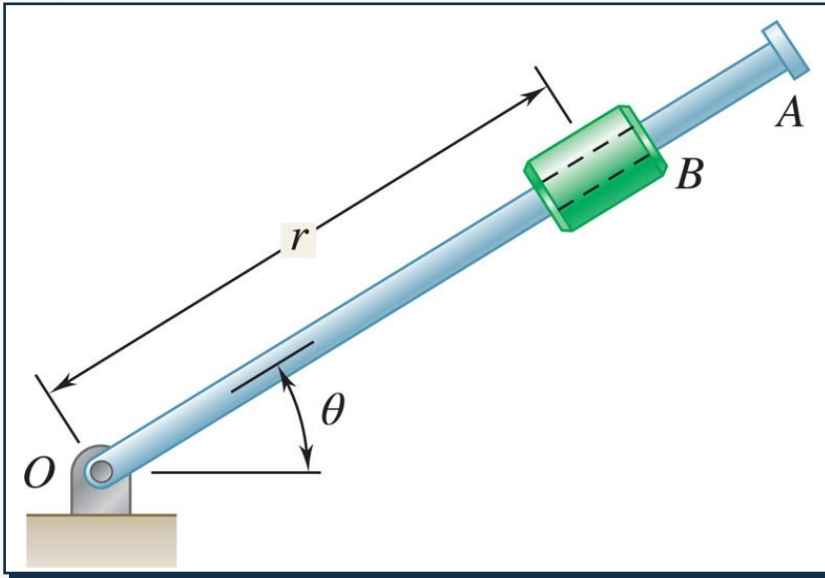


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Free Body Diagrams and Kinetic Diagrams

⚙️ Draw the FBD and KD for the collar B.

- Assume there is friction acting between the rod and collar, motion is in the vertical plane, and θ is increasing.

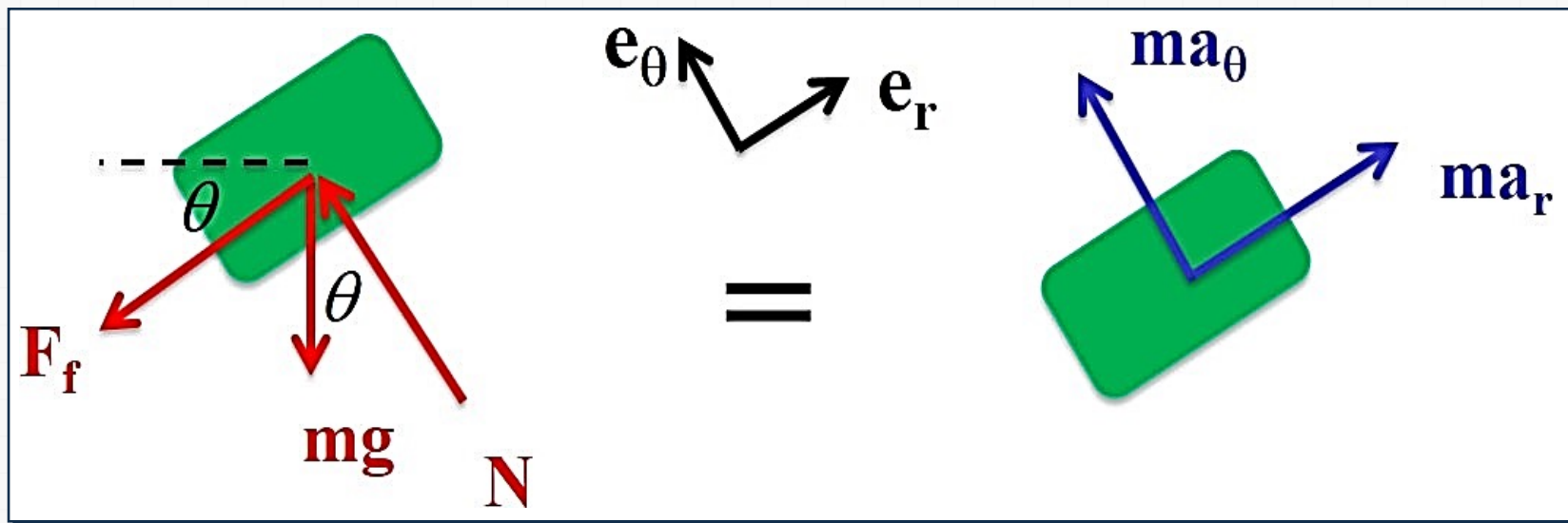


- Isolate body
- Axis
- Applied forces
- Replace supports with forces
- Dimensions
- Kinetic diagram

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Free Body Diagrams and Kinetic Diagrams

⚙ Draw the FBD and KD for the collar B.



4 Kinetics of Particles: Newton's Second Law

Kinetics: Normal and Tangential Coordinates

Aircraft and roller coasters can both **experience large normal forces** during turns.



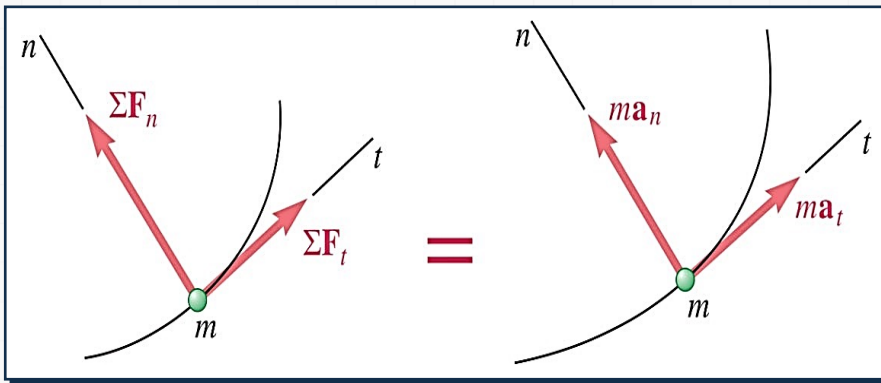
4 Kinetics of Particles: Newton's Second Law

Equations of Motion

⚙ Newton's second law

$$\Sigma F = ma$$

- For tangential and normal components,



$$\Sigma F_t = ma_t$$

$$\Sigma F_n = ma_n$$

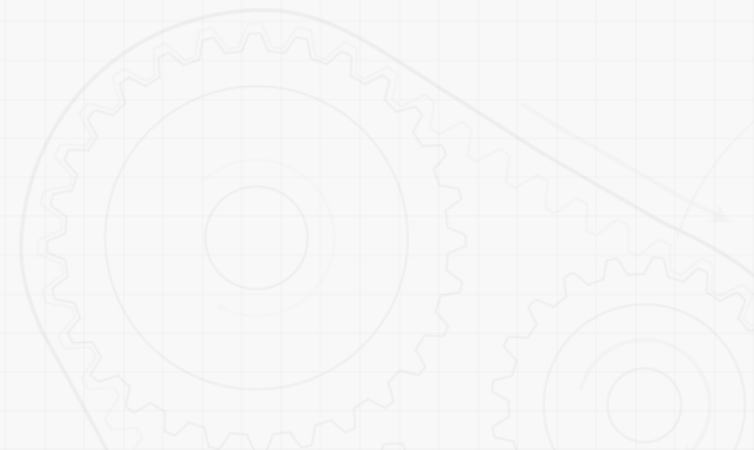
$$\Sigma F_t = m \frac{dv}{dt}$$

$$\Sigma F_n = m \frac{v^2}{\rho}$$

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Kinetics: Normal and Tangential Coordinates

Hydraulic actuators, extending robotic arms, and centrifuges as shown below are often analyzed using **radial and transverse coordinates**.



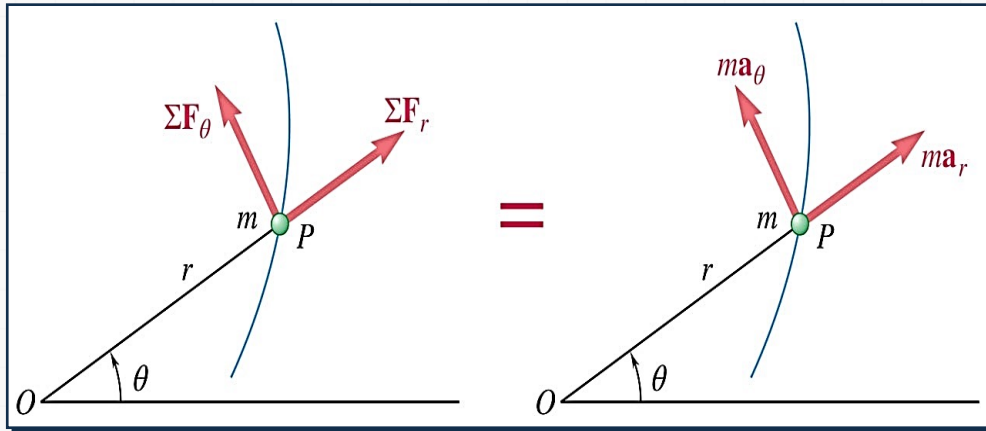
4 Kinetics of Particles: Newton's Second Law

Eqs of Motion in Radial & Transverse Components

⚙ Consider particle at r and θ , in polar coordinates,

$$\Sigma F_r = ma_r = m(\ddot{r} - r\dot{\theta}^2)$$

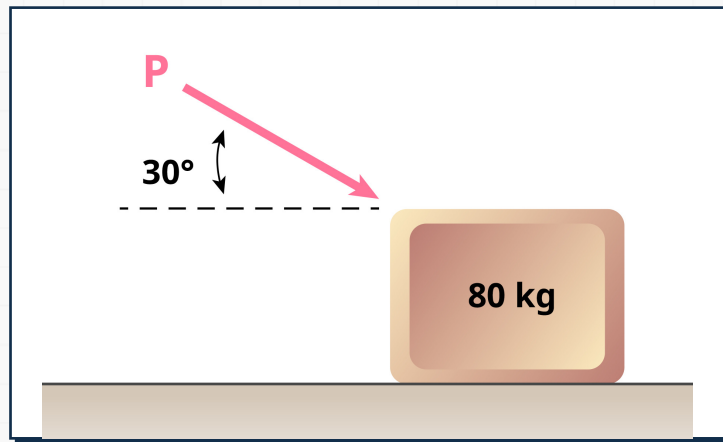
$$\Sigma F_\theta = ma_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$



4 Kinetics of Particles: Newton's Second Law

Sample Problem 1

⚙️ A 80-kg block rests on a horizontal plane.



- Find the magnitude of the force P required to give the block an acceleration of 2.5 m/s^2 to the right. The coefficient of kinetic friction between the block and plane is $\mu_k = 0.25$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 1

⚙️ **A 80-kg block rests on a horizontal plane.**

• **Strategy**

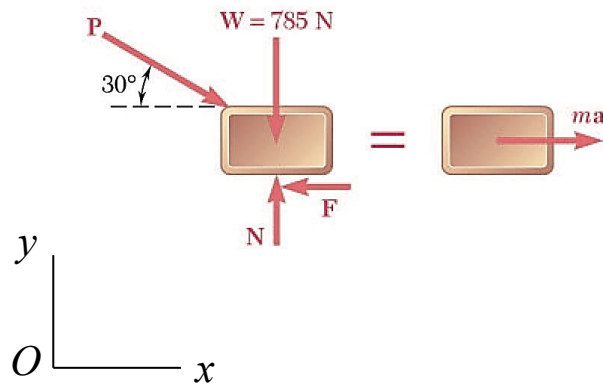
- Resolve the equation of motion for the block into two rectangular component equations.
- Unknowns consist of the applied force $\mathbf{P}$ and the normal reaction $\mathbf{N}$ from the plane. The two equations may be solved for these unknowns.

4 Kinetics of Particles: Newton's Second Law

Sample Problem 1

⚙️ Modeling and Analysis

$$W = (80\text{kg})\left(9.81\frac{\text{m}}{\text{s}^2}\right) = 785\text{ N}$$



$$m = 80\text{kg}$$

$$F = \mu_k N = 0.25N$$

- Resolve the equation of motion for the block into two rectangular component equations.

$$\sum F_x = ma :$$

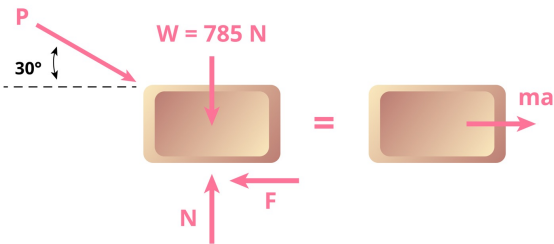
$$P \cos 30^\circ - 0.25N = (80\text{kg})(2.5\text{m/s}^2) \\ = 200$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 1

⚙️ Modeling and Analysis

$$W = (80\text{ kg})\left(9.81\frac{\text{m}}{\text{s}^2}\right) = 785\text{ N}$$



$$m = 80\text{ kg}$$

$$F = \mu_k N = 0.25N$$

- Resolve the equation of motion for the block into two rectangular component equations.

$$\sum F_y = 0 :$$

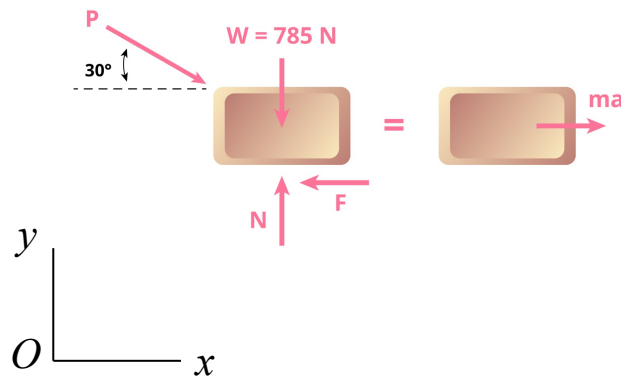
$$N - P \sin 30^\circ - 785 = 0$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 1

⚙ Modeling and Analysis

$$W = (80 \text{ kg}) \left(9.81 \frac{\text{m}}{\text{s}^2} \right) = 785 \text{ N}$$



$$m = 80 \text{ kg}$$

$$F = \mu_k N = 0.25N$$

- Unknowns consist of the applied force P and the normal reaction N from the plane.
 - The two equations may be solved for these unknowns.

$$N = P \sin 30^\circ + 785$$

$$P \cos 30^\circ - 0.25(P \sin 30^\circ + 785) = 200$$

$$P = 535 \text{ N}$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 1

⚙️ Reflect and Think

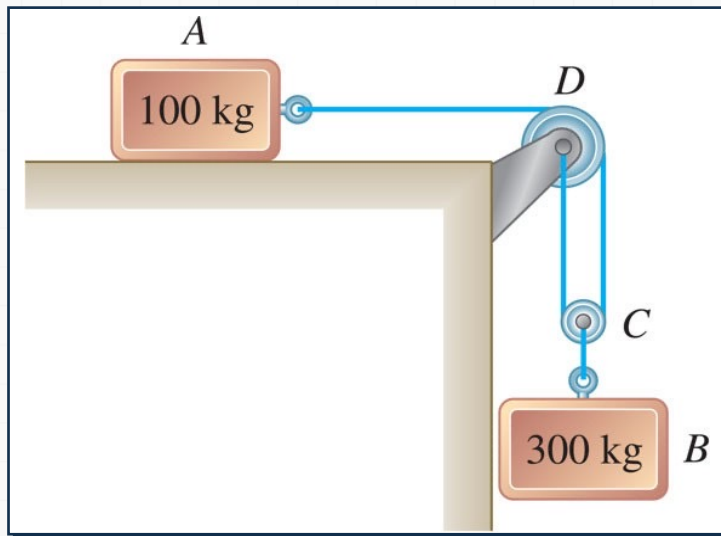
When you begin pushing on an object, you first have to overcome the static friction force ($F = \mu_s N$) before the object will move.

Also note that the downward component of force P increases the normal force N , which in turn increases the friction force F that you must overcome.

4 Kinetics of Particles: Newton's Second Law

Sample Problem 2

⚙ The two blocks shown start from rest.



• Determine

- The horizontal plane and the pulley are frictionless, and the pulley is assumed to be of negligible mass.
- Determine the acceleration of each block and the tension in the cord.

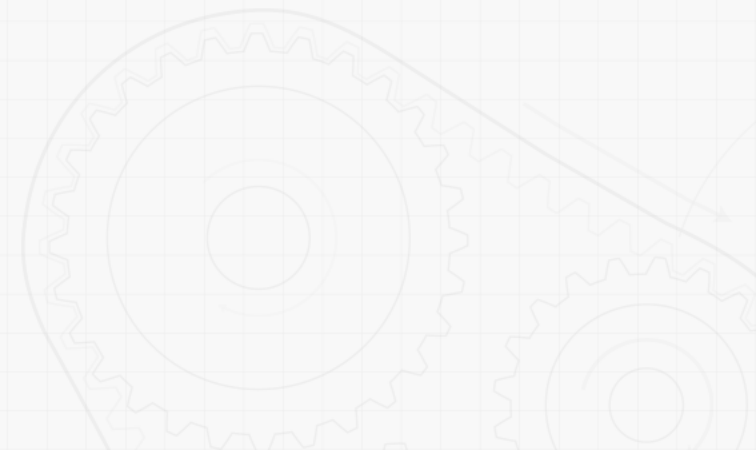
4 Kinetics of Particles: Newton's Second Law

Sample Problem 2

⚙️ **The two blocks shown start from rest.**

• **Strategy**

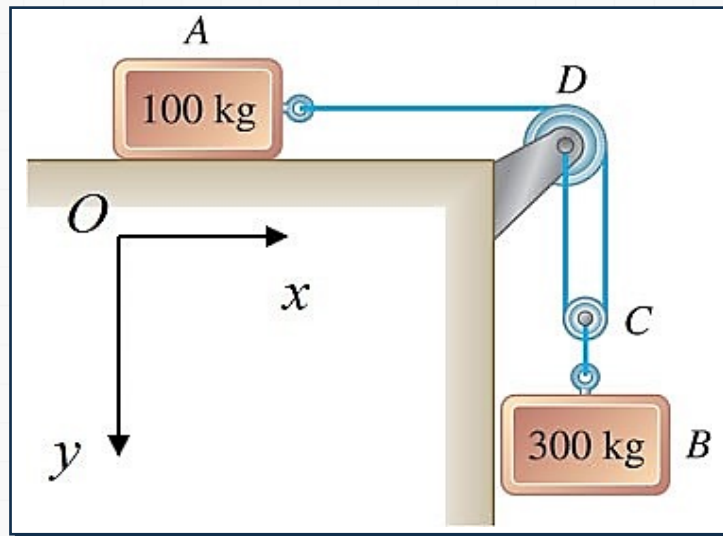
- Write the kinematic relationships for **the dependent motions** and **accelerations of the blocks**.
- Write the equations of motion for **the blocks and pulley**.
- Combine the kinematic relationships with the equations of motion to solve for the accelerations and cord tension.



4 Kinetics of Particles: Newton's Second Law

Sample Problem 2

⚙ Modeling and Analysis



- Write the kinematic relationships for the dependent motions and accelerations of the blocks.

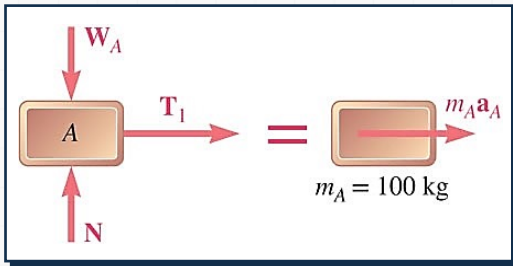
$$y_B = \frac{1}{2} x_A$$

$$a_B = \frac{1}{2} a_A$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 2

⚙ Modeling and Analysis



- Write equations of motion for the blocks and pulley.

$$\sum F_x = m_A a_A :$$

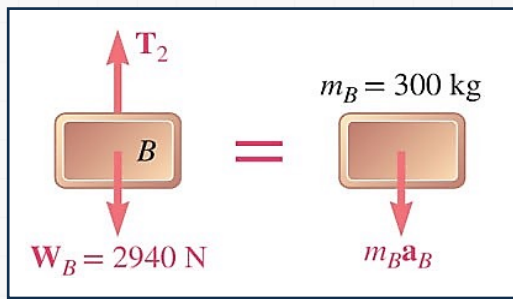
$$T_1 = (100\text{kg})a_A$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 2

⚙ Modeling and Analysis

$$\sum F_y = m_B a_B :$$



- Write equations of motion for **the blocks and pulley**.

$$\sum F_x = m_A a_A :$$

$$T_1 = (100 \text{ kg}) a_A$$

$$\sum F_y = m_B a_B :$$

$$m_B g - T_2 = m_B a_B$$

$$(300 \text{ kg})(9.81 \text{ m/s}^2) - T_2 = (300 \text{ kg}) a_B$$

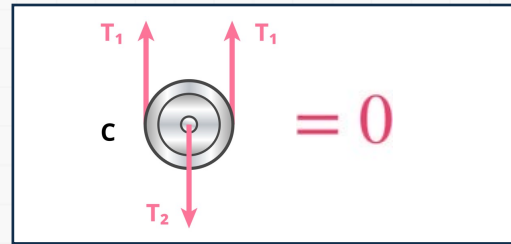
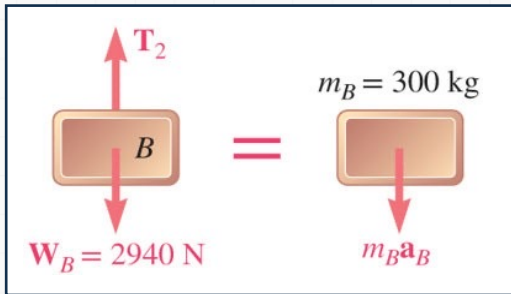
$$T_2 = 2940 \text{ N} - (300 \text{ kg}) a_B$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 2

⚙ Modeling and Analysis

$$\sum F_y = m_B a_B :$$



- Write equations of motion for the blocks and pulley.

$$\sum F_x = m_A a_A :$$

$$T_1 = (100 \text{ kg}) a_A$$

$$\sum F_y = m_B a_B :$$

$$m_B g - T_2 = m_B a_B$$

$$(300 \text{ kg})(9.81 \text{ m/s}^2) - T_2 = (300 \text{ kg}) a_B$$

$$T_2 = 2940 \text{ N} - (300 \text{ kg}) a_B$$

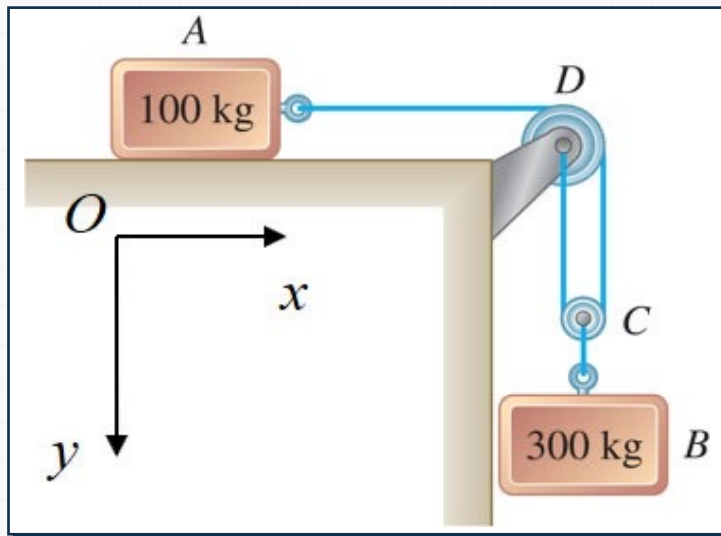
$$\sum F_y = m_c a_c = 0 :$$

$$T_2 - 2T_1 = 0$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 2

⚙ Modeling and Analysis



- Combine the kinematic relationships with the equations of motion to solve for the accelerations and cord tension.

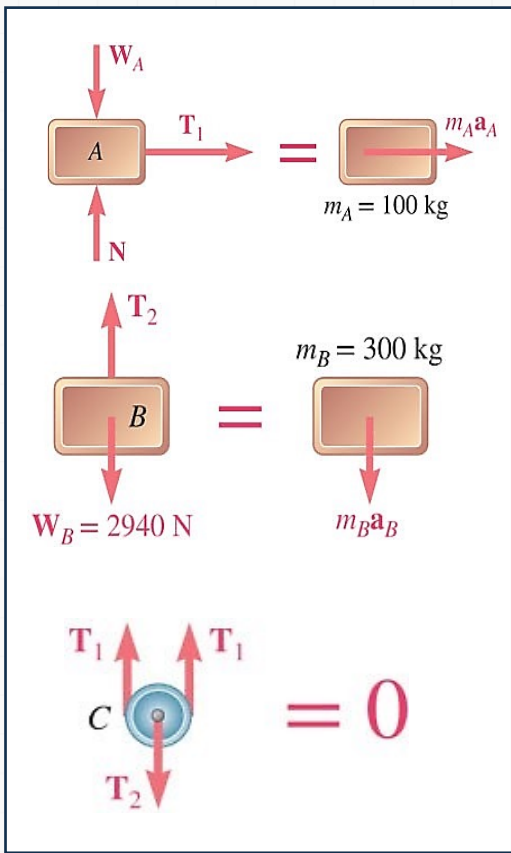
$$y_B = \frac{1}{2} x_A$$

$$a_B = \frac{1}{2} a_A$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 2

Modeling and Analysis



$$T_1 = (100 \text{ kg})a_A$$

$$\begin{aligned} T_2 &= 2940 \text{ N} - (300 \text{ kg})a_B \\ &= 2940 \text{ N} - (300 \text{ kg})\left(\frac{1}{2}a_A\right) \end{aligned}$$

$$T_2 - 2T_1 = 0$$

$$2940 \text{ N} - (150 \text{ kg})a_A - 2(100 \text{ kg})a_A = 0$$

$$a_A = 8.40 \text{ m/s}^2$$

$$a_B = \frac{1}{2}a_A = 4.20 \text{ m/s}^2$$

$$T_1 = (100 \text{ kg})a_A = 840 \text{ N}$$

$$T_2 - 2T_1 = 1680 \text{ N}$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 2

⚙️ Reflect and Think

Note that the value obtained for T_2 is not equal to the weight of block B .

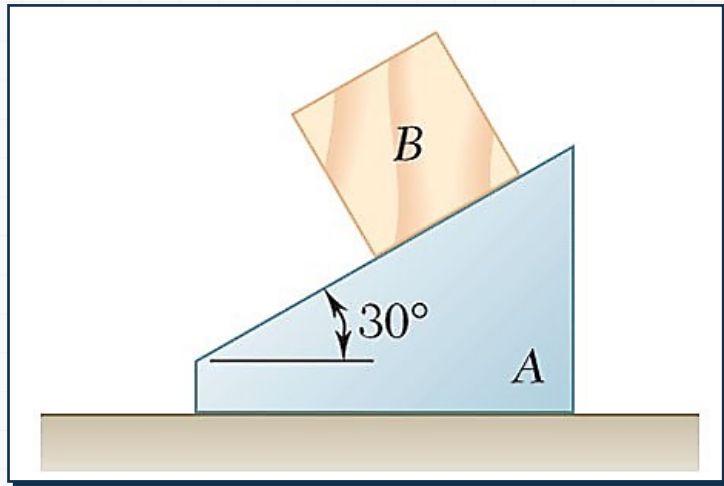
Rather than choosing B and the pulley as separate systems, you could have chosen the system to be B and the pulley.

In this case, T_2 would have been an internal force.

4 Kinetics of Particles: Newton's Second Law

Sample Problem 3

⚙ The 6-kg block B starts from rest and slides on the 15-kg wedge A, which is supported by a horizontal surface.



- Neglecting friction, determine
 - (a) the acceleration of the wedge,
 - (b) the acceleration of the block relative to the wedge.

4 Kinetics of Particles: Newton's Second Law

Sample Problem 3

⚙️ **The 6-kg block B starts from rest and slides on the 15-kg wedge A, which is supported by a horizontal surface.**

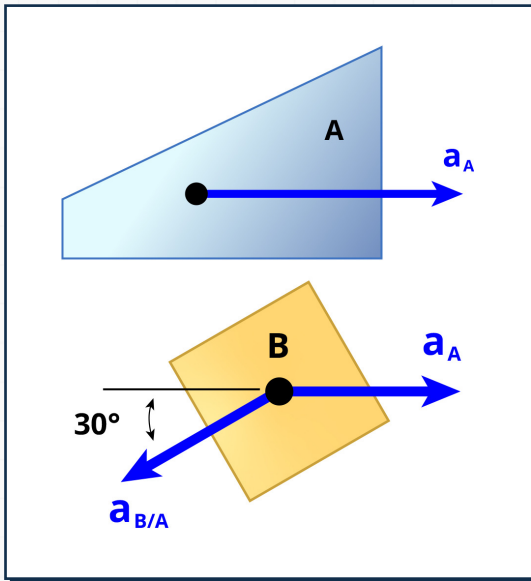
• **Strategy**

- The block is constrained to slide down the wedge. Therefore, their motions are dependent. Express the acceleration of block as the acceleration of wedge plus the acceleration of the block relative to the wedge.
- Write the equations of motion for the wedge and block.
- Solve for the accelerations.

4 Kinetics of Particles: Newton's Second Law

Sample Problem 3

⚙️ Modeling and Analysis



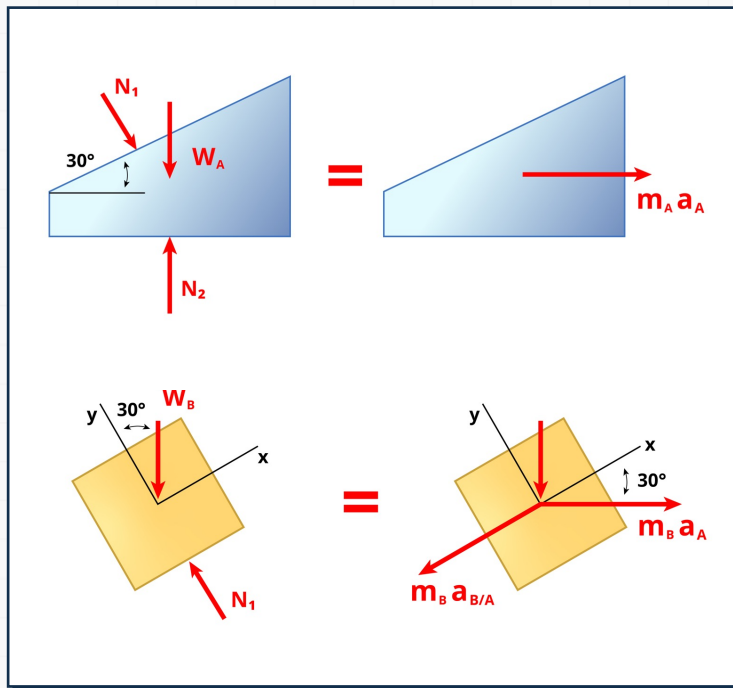
- The block is constrained to slide down the wedge.
- Therefore, their motions are dependent

$$\vec{a}_B = \vec{a}_A + \vec{a}_{B/A}$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 3

Modeling and Analysis



- Write equations of motion for wedge and block.

$$\sum F_x = m_A a_A :$$

$$N_1 \sin 30^\circ = m_A a_A$$

$$0.5 N_1 = (m_A) a_A$$

$$\sum F_x = m_B a_x = m_B (a_A \cos 30^\circ - a_{B/A}) :$$

$$- m_B g \sin 30^\circ$$

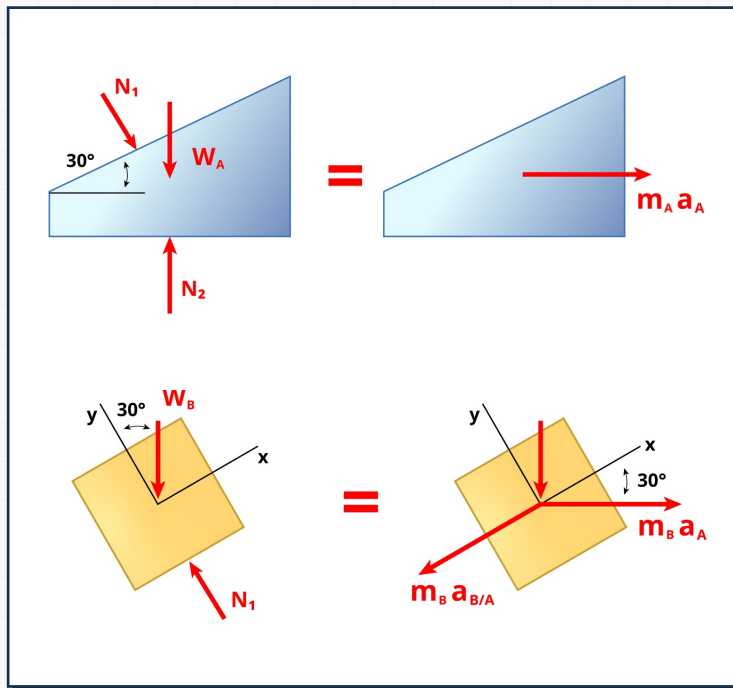
$$= (m_B) (a_A \cos 30^\circ - a_{B/A})$$

$$a_{B/A} = a_A \cos 30^\circ + g \sin 30^\circ$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 3

Modeling and Analysis



- Write equations of motion for wedge and block.

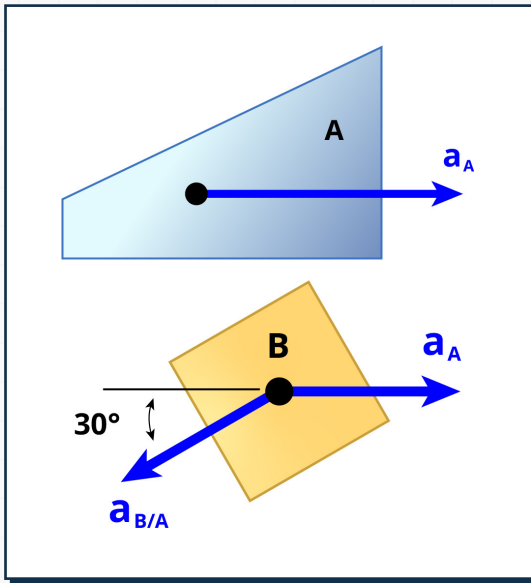
$$\sum F_y = m_B a_y = m_B (-a_A \sin 30^\circ):$$

$$N_1 - m_B g \cos 30^\circ = -(m_B)(a_A \sin 30^\circ)$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 3

⚙ Modeling and Analysis



- Solve for the accelerations.

$$0.5N_1 = (m_A)a_A$$

$$N_1 - m_B g \cos 30^\circ = -(m_B)a_A \sin 30^\circ$$

$$2(m_A)a_A - m_B g \cos 30^\circ = -(m_B)a_A \sin 30^\circ$$

$$a_A = \frac{gm_B \cos 30^\circ}{2m_A + m_B \sin 30^\circ}$$

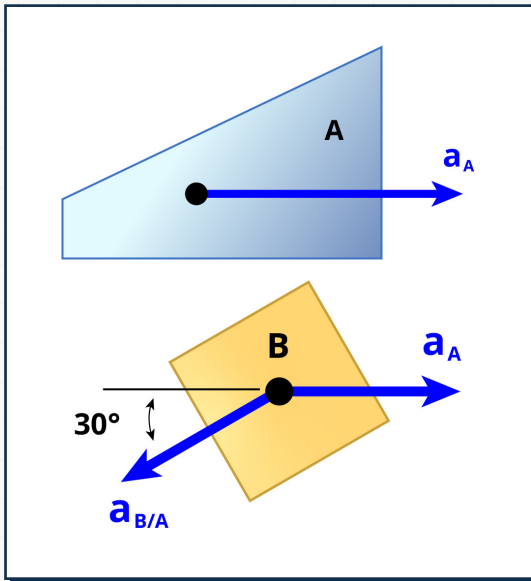
$$a_A = \frac{(6\text{ kg})(9.81\text{ m/s}^2) \cos 30^\circ}{2(15\text{ kg}) + (6\text{ kg}) \sin 30^\circ}$$

$$a_A = 1.545\text{ m/s}^2$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 3

⚙ Modeling and Analysis



- Solve for the accelerations.

$$a_{B/A} = a_A \cos 30^\circ + g \sin 30^\circ$$

$$a_{B/A} = (1.545 \text{ m/s}^2) \cos 30^\circ + (9.81 \text{ m/s}^2) \sin 30^\circ$$

$$a_{B/A} = 6.24 \text{ m/s}^2$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 3

⚙️ Reflect and Think

Many students are tempted to draw the acceleration of block B down the incline in the kinetic diagram.

It is important to recognize that this is the direction of the relative acceleration.

4 Kinetics of Particles: Newton's Second Law

Sample Problem 3

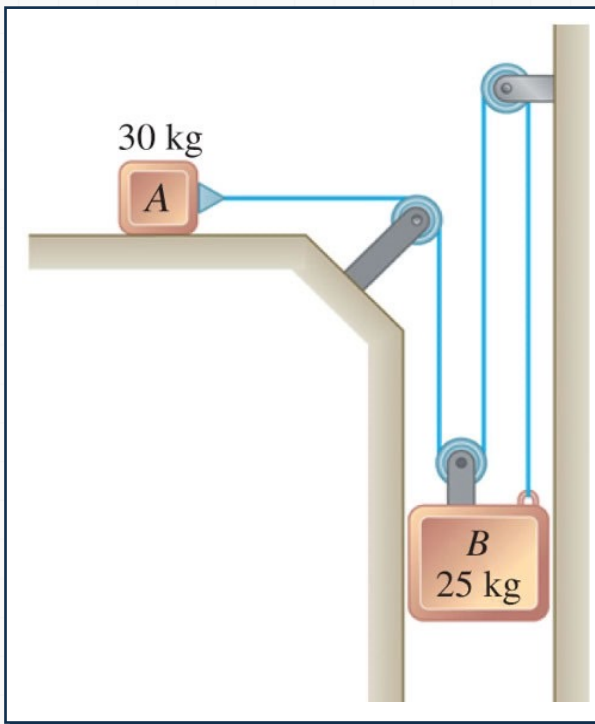
⚙️ Reflect and Think

Rather than the kinetic diagram you used for block B , you could have simply put unknown accelerations in the x and y directions and then used your relative motion equation to obtain more scalar equations.

4 Kinetics of Particles: Newton's Second Law

Group Problem Solving

⚙️ The two blocks shown are originally at rest.



- Determine

- Neglecting the masses of the pulleys and the effect of friction in the pulleys and between block A and the horizontal surface, determine
 - (a) the acceleration of each block,
 - (b) the tension in the cable.

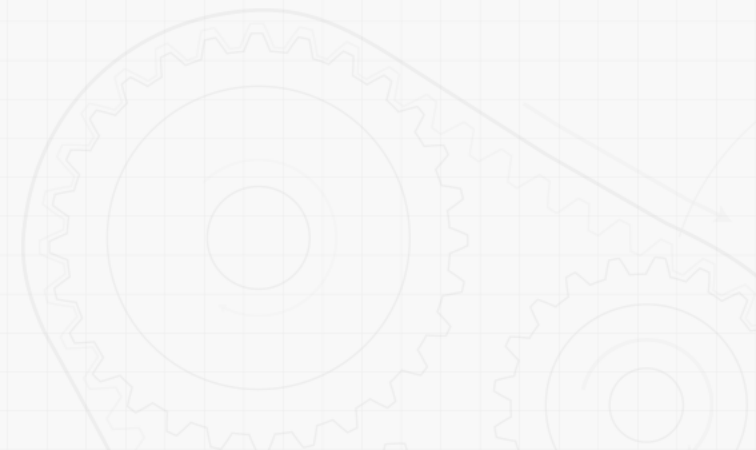
4 Kinetics of Particles: Newton's Second Law

Group Problem Solving

⚙️ **The two blocks shown are originally at rest.**

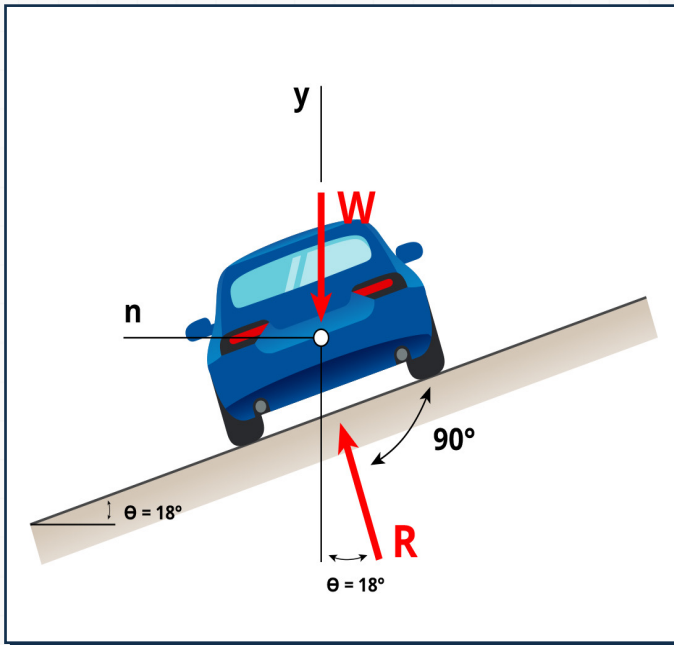
• **Strategy**

- Write the kinematic relationships for the dependent motions and accelerations of the blocks
- Draw the FBD and KD for each block
- Write the equations of motion for the blocks and pulley
- Combine the kinematic relationships with the equations of motion to solve for the accelerations and cord tension



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Sample Problem 4



• Determine

- Determine the rated speed of a highway curve of radius $\rho = 120m$ banked through an angle $\theta = 18^\circ$.
- The rated speed of a banked highway curve is the speed at which a car should travel if no lateral friction force is to be exerted at its wheels.

4 Kinetics of Particles: Newton's Second Law

Sample Problem 4



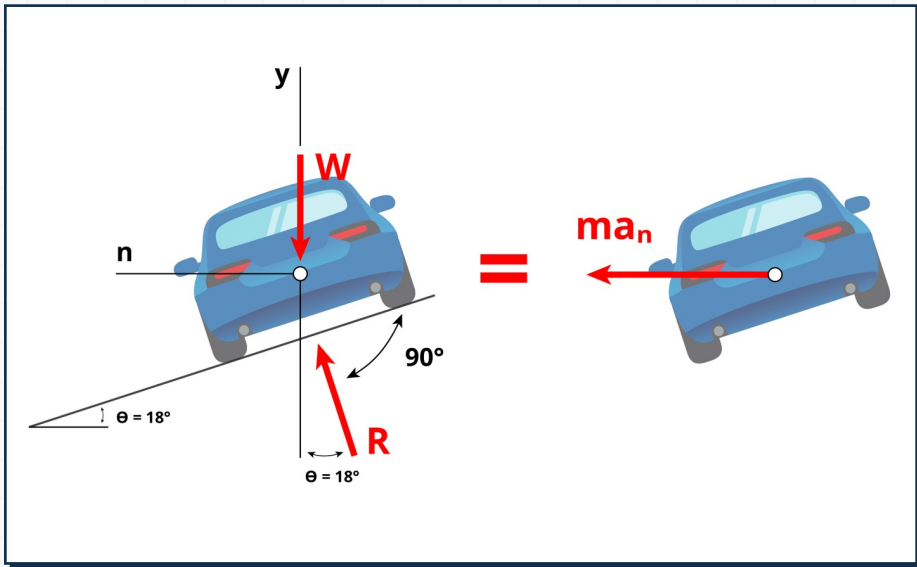
Strategy

- The car travels in a horizontal circular path with a normal component of acceleration directed toward the center of the path. The forces acting on the car are its **weight and a normal reaction from the road surface**.
- Resolve the equation of motion for the car into **vertical and normal components**.
- Solve for **the vehicle speed**.

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Sample Problem 4

⚙️ Modeling and Analysis



- Resolve the equation of motion for the car into **vertical and normal components**.

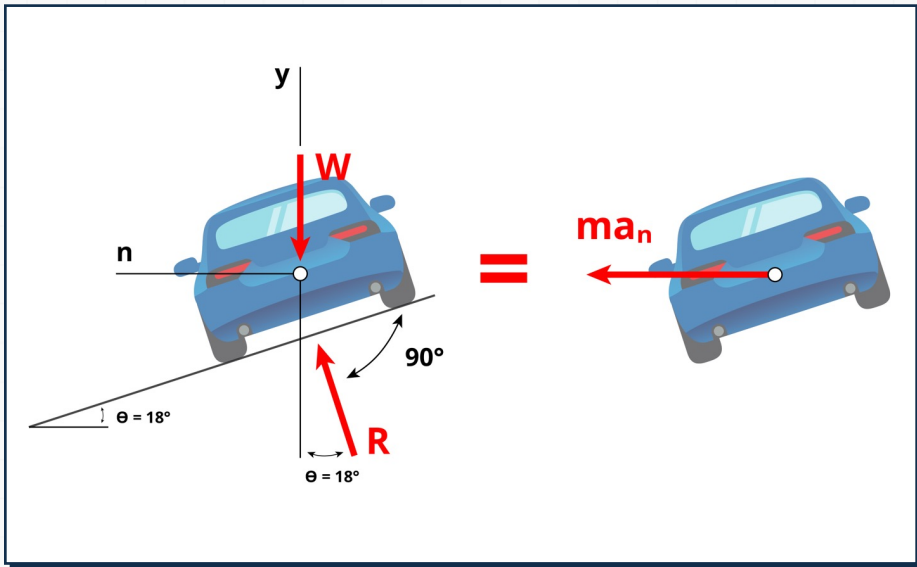
$$\sum F_y = 0 : \quad R \cos \theta - W = 0$$

$$R = \frac{W}{\cos \theta}$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 4

⚙️ Modeling and Analysis



- Resolve the equation of motion for the car into **vertical and normal components**.

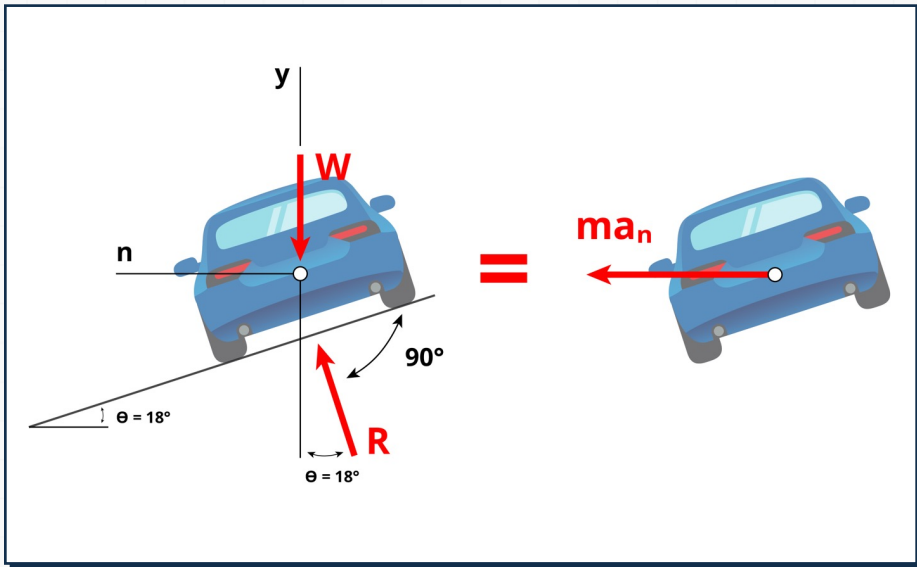
$$\sum F_n = ma_n: R \sin \theta = \frac{W}{g} a_n$$

$$\frac{W}{\cos \theta} \sin \theta = \frac{W}{g} \frac{v^2}{\rho}$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 4

⚙ Modeling and Analysis



- Solve for the vehicle speed.

$$\begin{aligned} v^2 &= g\rho \tan\theta \\ &= (9.81 \text{ m/s}^2)(120 \text{ m}) \tan 18^\circ \end{aligned}$$

$$v = 19.56 \text{ m/s} = 70.4 \text{ km/h}$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 4

⚙️ Reflect and Think

For a highway curve, this seems like a reasonable speed for avoiding a spin-out. If the roadway were banked at a larger angle, would the rated speed be larger or smaller than this calculated value?

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Sample Problem 4

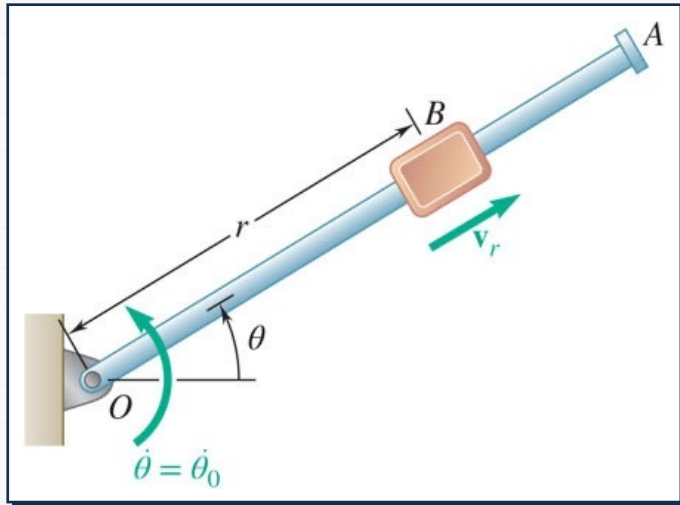
⚙️ Reflect and Think

For this problem, the tangential direction is into the page; since you were not asked about forces or accelerations in this direction, you did not need to analyze motion in the tangential direction.

4 Kinetics of Particles: Newton's Second Law

Sample Problem 5

⚙️ A block B of mass m can slide freely on a frictionless arm OA which rotates in a horizontal plane at a constant rate $\dot{\theta}_0$.



• Determine

- Knowing that B is released at a distance r_0 from O , express as a function of r .
 - a) the component v_r of the velocity of B along OA , and
 - b) the magnitude of the horizontal force exerted on B by the arm OA .

4 Kinetics of Particles: Newton's Second Law

Sample Problem 5

⚙️ **A block B of mass m can slide freely on a frictionless arm OA which rotates in a horizontal plane at a constant rate $\dot{\theta}_0$.**

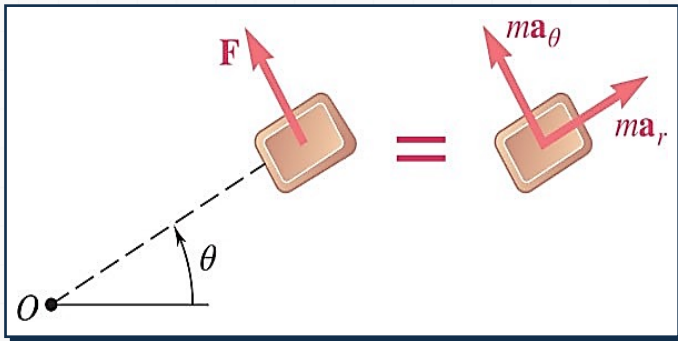
• **Strategy**

- Write the radial and transverse equations of motion for the block.
- Integrate the radial equation to find an expression for the radial velocity.
- Substitute known information into the transverse equation to find an expression for the force on the block.

4 Kinetics of Particles: Newton's Second Law

Sample Problem 5

⚙ Modeling and Analysis



- Write the radial and transverse equations of motion for the block.

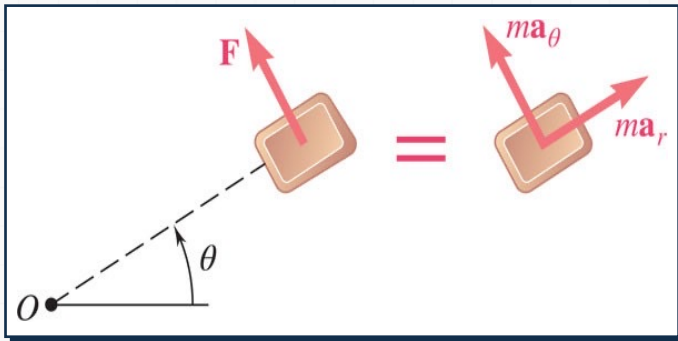
$$\sum F_r = ma_r: \quad 0 = m(\ddot{r} - r\dot{\theta}^2)$$

$$\sum F_\theta = ma_\theta: \quad F = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 5

⚙ Modeling and Analysis



- Integrate the radial equation to find an expression for the radial velocity.

$$\ddot{r} = \dot{v}_r = \frac{dv_r}{dt} = \frac{dv_r}{dr} \frac{dr}{dt} = v_r \frac{dv_r}{dr}$$

$$v_r dv_r = r \dot{\theta}^2 dr = r \dot{\theta}_0^2 dr$$

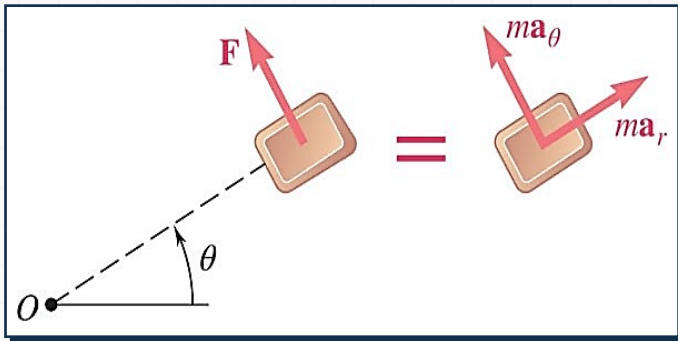
$$\int_0^{v_r} v_r dv_r = \dot{\theta}_0^2 \int_{r_0}^r r dr$$

$$v_r^2 = \dot{\theta}_0^2 (r^2 - r_0^2)$$

4 Kinetics of Particles: Newton's Second Law

Sample Problem 5

⚙ Modeling and Analysis



- Substitute known information into the transverse equation to find **an expression for the force on the block.**

$$F = 2m\dot{\theta}_0^2 (r^2 - r_0^2)^{1/2}$$

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Sample Problem 5

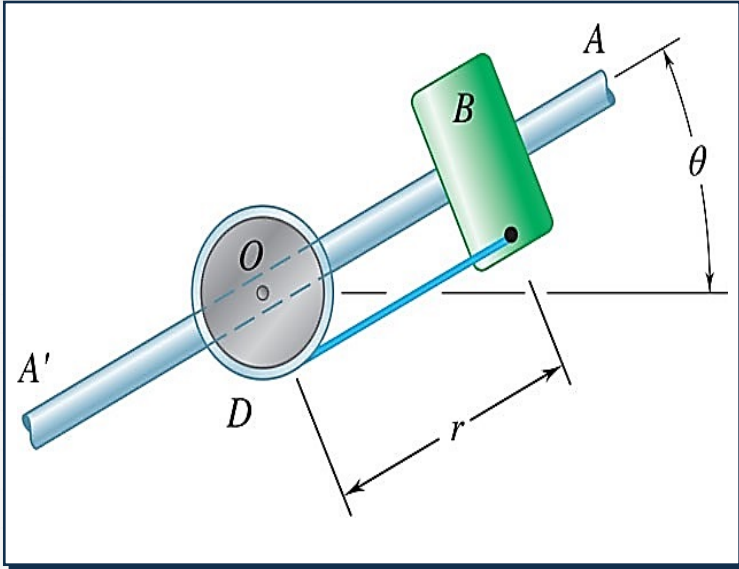
⚙️ Reflect and Think

Introducing radial and transverse components of force and acceleration involves using components of velocity as well in the computations. But this is still much simpler and more direct than trying to use other coordinate systems.

Even though the radial acceleration is zero, the collar accelerates relative to the rod with acceleration $\ddot{r}$.

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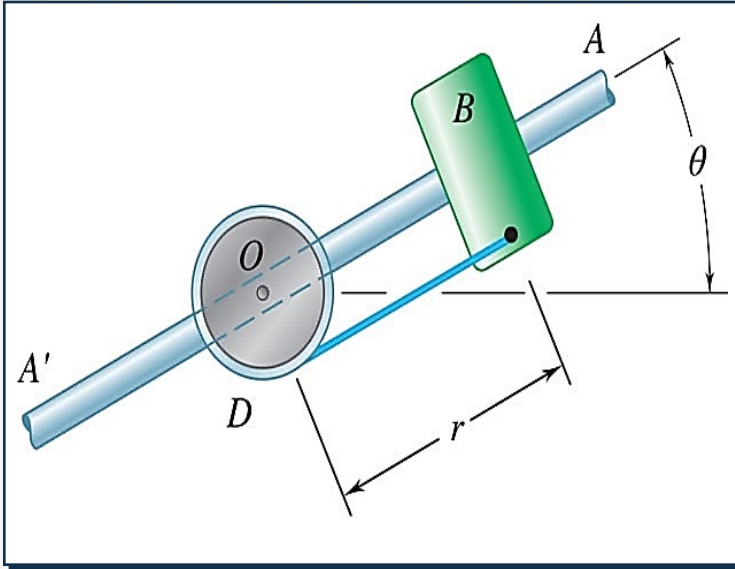
Group Problem Solving



The 3-kg collar B slides on the frictionless arm AA' . The arm is attached to drum D and rotates about O in a horizontal plane at the rate where and t are expressed in rad/s and seconds, respectively.

4 Kinetics of Particles: Newton's Second Law

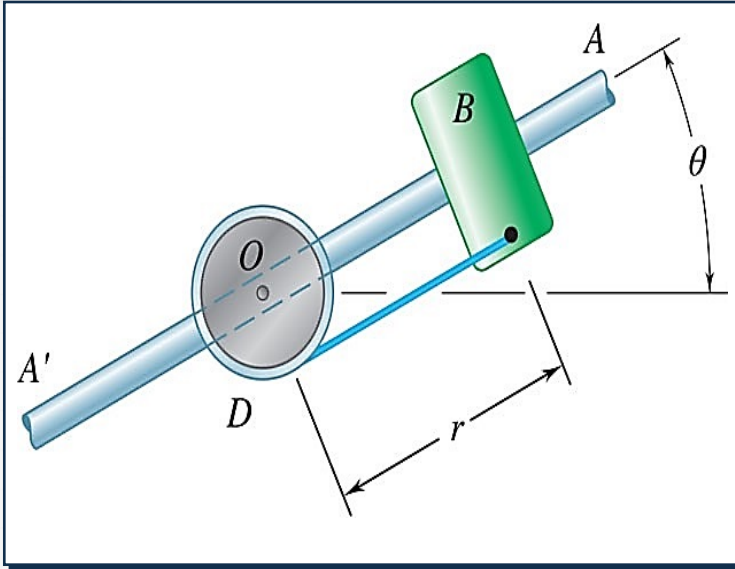
Group Problem Solving



As the arm-drum assembly rotates, a mechanism within the drum releases the cord so that the collar moves outward from O with a constant speed of 0.5 m/s .

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Group Problem Solving

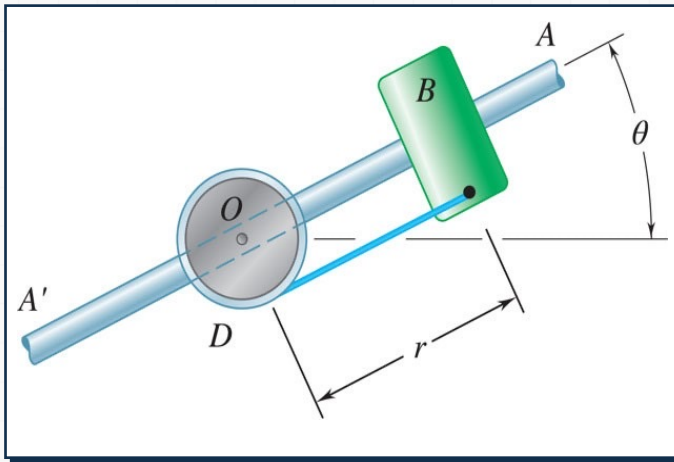


Knowing that at $t = 0, r = 0$, determine the time at which the tension in the cord is equal to the magnitude of the horizontal force exerted on B by arm AA' .

4 Kinetics of Particles: Newton's Second Law

Group Problem Solving

⚙ Strategy



- Draw the **FBD** and **KD** for the collar.
- Write **the equations of motion** for the collar.
- Determine **kinematics** of the collar.
- Combine **the equations of motion** with **kinematic relationships** and solve.

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References

- Beer, Johnston, Cornwell, Self, Sanghi, "Vector Mechanics for Engineers: Dynamics" 12th Edition, Chapter 12